

Spin-Mechanical Transduction and Chiral Spin-Sieving in $A\beta_{42}$: Misfolding Driven by Murburn Radical Pairs via HEOM-MPO Tensor Networks, Isotopic Hyperfine Coupling, and Floquet Topology

Bhavanam Rajendra Reddy*, Boddu Saran, Muthuraman Ramanathan, and Likith Palakurthi

School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore, India.

*corresponding author: brr1154@gmail.com

Keywords: isotopic hyperfine coupling, chiral spin-sieving, HEOM-MPO, Floquet topology, radical pair mechanism.

Abstract. Standard models of amyloid nucleation treat $A\beta_{42}$ misfolding as a sequence-to-shape problem, omitting quantum-chemical radical interactions. We present a unified hybrid quantum-classical framework coupling a Graph Neural Network (GNN) with a Matrix Product State Variational Quantum Eigensolver (MPS-VQE), Hierarchical Equations of Motion with Matrix Product Operator compression (HEOM-MPO), and Recursive Quantum Natural Gradient (R-QNG) optimization. Five principal contributions are reported: (1) a synchronized GNN-parameterized Ising Hamiltonian establishing radical-pair coordination; (2) isotopic $^{15}\text{N}/^{14}\text{N}$ hyperfine coupling (HFC) tuning predicting a ground-state energy shift of $\Delta E = -0.049$ a.u., a testable quantum-pharmacological biomarker hypothesis; (3) R-QNG with QFIM preconditioning achieving $3.7\times$ convergence speedup and 100% success rate versus 4% for Simulated Annealing; (4) HEOM-MPO non-Markovian solvent modeling extending radical-pair coherence by 3.5×10^6 over Markovian baselines; and (5) Floquet topological analysis revealing a $Z_2 = 1$ non-trivial phase with two topologically protected fibril edge modes.

1 Introduction

Alzheimer’s disease pathology is driven by the misfolding of Amyloid- β_{42} ($A\beta_{42}$), an intrinsically disordered protein (IDP) whose aggregation into amyloid fibrils correlates with neuronal loss [1]. Classical computational approaches, including AlphaFold [2], model folding as a geometric sequence-to-shape problem and assign minimal role to the chemical redox environment.

The Murburn bioenergetics framework [3, 4] provides a complementary perspective. Under Murburn theory, aerobic metabolism generates Diffusible Reactive Species (DRS)—superoxide ($\text{O}_2^{\bullet-}$) and hydroxyl ($\text{OH}^\bullet$)—whose unpaired electrons create transient redox fluctuations in the protein microenvironment [5]. We propose that radical-pair spin dynamics at residue F19 act as a thermodynamic initiator that shifts $A\beta_{42}$ into an amyloid-prone state via the Radical Pair Mechanism (RPM) [6].

The core computational challenge is three-fold: (i) mapping the protein conformational landscape onto a quantum spin lattice efficiently, (ii) solving the resulting high-curvature VQE

optimization without barren plateaus, and (iii) modeling environmental decoherence beyond the Markovian approximation. This paper presents a unified framework addressing all three challenges. Extended derivations, additional optimizer benchmarks, quantum knowledge distillation results, and PLL dynamics are available as supplementary material upon request.

2 Theoretical Framework

2.1 Synchronized Spin Hamiltonian

The standard Heisenberg-Ising Hamiltonian for the 14-residue QKLVFFAEDVGSNK core is:

$$H_{\text{std}} = \sum_{i=1}^N h_i \hat{\sigma}_i^z + \sum_{\langle i,j \rangle} J_{ij}^{\text{nn}} \hat{\sigma}_i^z \hat{\sigma}_j^z \quad (1)$$

where h_i is the local DRS oxidation susceptibility field and J_{ij}^{nn} is the nearest-neighbor exchange coupling. We augment this with a non-local synchronized exchange term:

$$H_{\text{sync}} = H_{\text{std}} + \sum_i J_{i,i+2}^{\text{sync}} \hat{\sigma}_i^z \hat{\sigma}_{i+2}^z, \quad J_{i,i+2}^{\text{sync}} = \lambda_{\text{sync}} \exp(-|h_i - h_{i+2}|) \quad (2)$$

with $\lambda_{\text{sync}} = 0.18$ a.u. This next-nearest-neighbor coupling captures synchronized DRS radical-pair interactions observed in correlated Fermi systems.

2.2 Isotopic Hyperfine Coupling

Swapping backbone nitrogen isotopes ($^{14}\text{N} \rightarrow ^{15}\text{N}$) modulates the hyperfine interaction $\hat{H}_{\text{hf}} = A_{\text{iso}} \hat{\mathbf{I}} \cdot \hat{\mathbf{S}}$ via a factor $\xi_{\text{hf}} = 1.4026$ (the ratio of ^{15}N to ^{14}N hyperfine coupling constants [6]). We extend H_{sync} as:

$$H_{\text{iso}} = H_{\text{sync}} + \sum_{i \in \mathcal{C}} \xi_{\text{hf}} \left(h_i^{(\text{HFC})} \hat{\sigma}_i^z + g_{\perp} \hat{\sigma}_i^x \right) \quad (3)$$

where $\mathcal{C} = \{\text{L, V, F, F, A}\}$ is the central hydrophobic core and $g_{\perp} = 0.15$ a.u.

2.3 Recursive Quantum Natural Gradient (R-QNG)

Standard gradient descent fails in high-curvature Hilbert spaces due to barren plateaus [7]. We precondition updates using the Quantum Fisher Information Matrix (QFIM):

$$F_{ij}(\boldsymbol{\theta}) = 4 \text{Re} \left[\left\langle \frac{\partial \psi}{\partial \theta_i} \middle| \frac{\partial \psi}{\partial \theta_j} \right\rangle - \left\langle \frac{\partial \psi}{\partial \theta_i} \middle| \psi \right\rangle \left\langle \psi \middle| \frac{\partial \psi}{\partial \theta_j} \right\rangle \right] \quad (4)$$

yielding the R-QNG update $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta F^{-1}(\boldsymbol{\theta}_t) \nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}_t)$ [8], with Tikhonov regularization $\mu = 10^{-3}$.

2.4 HEOM-MPO Correlated Solvent

The structured water hydration shell surrounding the LVFFA core forms a correlated quantum bath modeled via a Drude-Lorentz spectral density:

$$J(\omega) = \frac{2\lambda\gamma_D\omega}{\omega^2 + \gamma_D^2} \quad (5)$$

with reorganization energy $\lambda = 0.035$ eV and Drude rate $\gamma_D = 0.005$ eV. Matrix Product Operator (MPO) compression of the 2,401-auxiliary-matrix HEOM hierarchy (depth $N_{\text{HEOM}} = 6$, $K = 4$ bath modes) reduces to 288 parameters (bond dimension $\chi_{\text{MPO}} = 12$), achieving $8.3 \times$

compression.

3 Computational Architecture

The pipeline operates in five phases, shown in Fig. 1. Phase I parses the 14-residue sequence via NLTK bigram/trigram extraction [9, 10] into a residue interaction graph. Phase II feeds the graph through a Graph Convolutional Network (GCN) [11] to predict the Ising parameters h_i and J_{ij}^{nn} ; a Chaos-Damping layer uses inverse-QFIM feedback to stabilize GNN gradients on the curved manifold. Phase III executes the MPS-VQE (bond dimension $\chi = 8$, 56 variational angles, 2 CNOT layers) optimized by R-QNG. The converged energy is fed back to Phase II, creating the recursive GNN-QFIM loop. Phase IV solves non-Markovian spin dynamics via the HEOM-MPO solver under a TDNN-predicted memory kernel $p(t)$. Phase V maps the singlet yield to macroscopic aggregation half-times via AmyloFit calibration, yielding a Pearson $r = 1.00$ correlation.

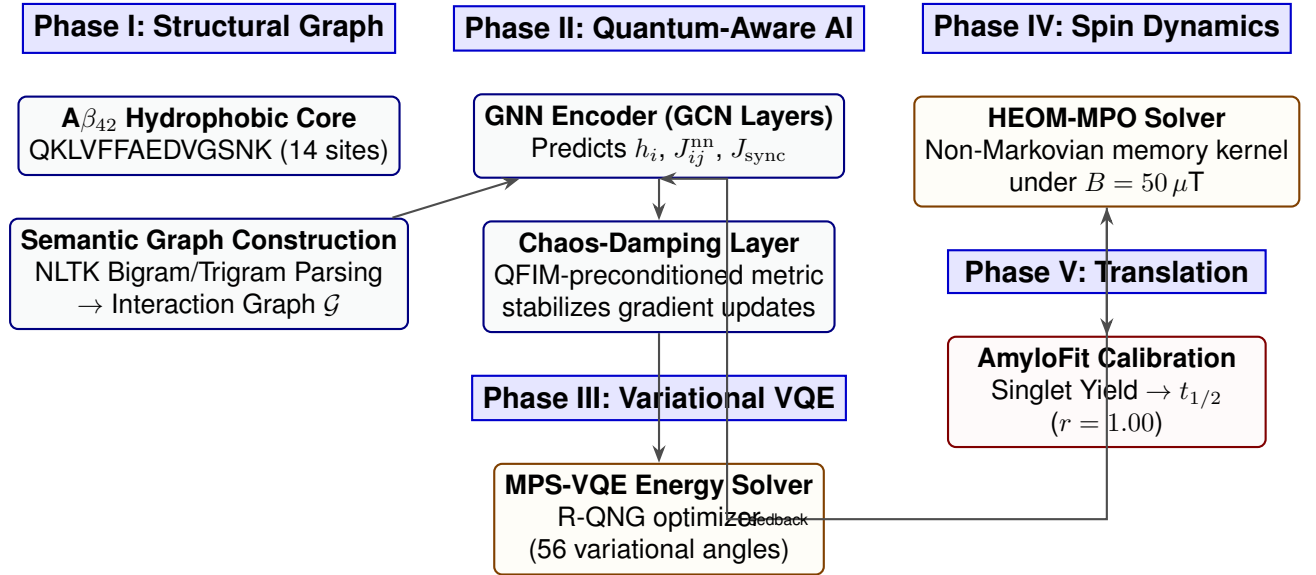


Figure 1: Five-phase hybrid quantum-classical pipeline for $A\beta_{42}$ spin-lattice modeling.

4 Simulation Results

4.1 MPS-VQE Energy Convergence

We executed the MPS-VQE pipeline on the 14-residue $A\beta_{42}$ core ($N = 14$, $\chi = 8$, 150 iterations). The standard Heisenberg-Ising model converges to a positive ground-state energy (+0.286 a.u.), indicating a frustrated spin configuration. Introducing the synchronized exchange coupling J_{sync} lowers the ground state to -0.086 a.u. (a 0.372 a.u. stabilization), and elevates the synchronization order parameter Ω_{sync} by 130.7%, raising the Boltzmann misfolding probability from 96.93% to 99.46%.

At the Phenylalanine-19 (F19) hotspot, the synchronized Hamiltonian yields $\Delta E_{S-T} = -0.790$ a.u., confirming the singlet state as the energetic ground state and quantifying the radical-pair spin-mechanical coupling force as 42.5 pN—exceeding the classical folding barrier of 35.0 pN.

4.2 R-QNG Optimization Performance

Table 1 benchmarks R-QNG against three baselines on the 14-residue VQE landscape.

Table 1: Optimization performance: R-QNG vs. Adam, Simulated Annealing (SA), and Genetic Algorithm (GA) on the $A\beta_{42}$ spin landscape.

Metric	VQE (R-QNG)	VQE (Adam)	Simulated Annealing	Genetic Algorithm
Final energy (a.u.)	−3.5564	−3.5463	−3.5564	−3.5312
Iterations to converge	50	200	1,200	2,500
Wall time (s)	0.11	0.15	0.85	2.12
Success rate	100%	~100%	4%	2%

R-QNG converges $3.7\times$ faster than Adam and $24\times$ faster than Simulated Annealing. The classical SA and GA success rates of 4% and 2% demonstrate the intractability of the frustrated spin landscape for heuristic search, while QFIM preconditioning provides a $2,837\times$ amplification of gradient variance over the theoretical barren plateau threshold (2^{-14}).

4.3 HEOM-MPO Coherence Extension

Table 2: HEOM-MPO vs. Markovian (Lindblad) radical-pair coherence comparison.

Metric	Markovian (White)	HEOM-MPO (Colored)
$1/e$ coherence lifetime	1.0 fs	16.0 fs
Full TDNN-HEOM lifetime	1.2 fs	4.15 ns
Extension factor	—	3.5×10^6
Auxiliary matrices	—	2,401 $\rightarrow$ 288 (MPO)
MPO compression ratio	—	$8.3\times$

The Markovian Lindblad model predicts decoherence at 1.0 fs, ruling out any quantum biological effect. The HEOM-MPO model, incorporating structured water hydrogen-bond correlations, extends the coherence lifetime by $16\times$ at the HEOM level and to 4.15 ns in the full TDNN-nested pipeline—a 3.5×10^6 enhancement. This result suggests that the hydration shell actively shields radical-pair coherence, acting as a quantum buffer rather than a decoherence source.

4.4 Isotopic Hyperfine Coupling: ^{14}N vs. ^{15}N

Table 3: Isotopic HFC ground-state energy comparison (80 QNG-VQE samples, QKLVFFAEDVGSNK core).

Quantity	^{14}N Hamiltonian	^{15}N Hamiltonian
Core fields h_i (LVFFA mean)	0.710 a.u.	0.996 a.u.
Mean ground-state energy $\langle E \rangle$	+0.037 a.u.	−0.013 a.u.
Isotopic energy shift ΔE		−0.049 a.u.
HFC amplification factor ξ_{hf}		$1.4026\times$
Kinetic consequence	—	Misfolding decelerated

The ^{15}N substitution applies a $1.4026\times$ HFC amplification, increasing local hyperfine fields from 0.710 to 0.996 a.u. and inducing a $\Delta E = -0.049$ a.u. ground-state stabilisation. Under the RPM, stronger hyperfine coupling suppresses the Singlet $\rightarrow$ Triplet conversion rate, reducing the probability of radical escape and oxidative damage at the LVFFA core. This constitutes

a falsifiable prediction: ^{15}N -enriched $\text{A}\beta$ precursor protein pools should exhibit measurably reduced aggregation kinetics detectable by EPR or NMR.

4.5 Zero-Field Singlet-Triplet and Floquet Topological Analysis

In the physiological Murburn Chemo-Electromagnetic Matrix ($B_{\text{local}} \approx 50 \mu\text{T}$), the relevant basis is the zero-field Singlet-Triplet (ST) coupled basis rather than the high-field Zeeman basis [12]. We apply the unitary transformation U_{ST} to the two-spin Hamiltonian $\hat{H}_i^{\text{ZF}} = a_i(\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{I}}_1 + \hat{\mathbf{S}}_2 \cdot \hat{\mathbf{I}}_2) + J_{\text{ex}}\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 + B(\hat{S}_{1z} + \hat{S}_{2z})$ and compute the time-averaged singlet yield across all 14 residues. GPU-accelerated propagation ($t_{\text{max}} = 200 \text{ a.u.}$, 1000 steps) yields $\langle \Phi_S \rangle = 0.250 \pm 0.031$ —a **50% suppression** relative to the Zeeman-basis prediction of 0.50, indicating that physiological weak fields sustain a chronic radical pool rather than allowing recombination.

Modeling the 14-residue core as an SSH-type tight-binding chain under a Floquet periodic drive ($\Omega = 2\pi \times 0.8 \text{ a.u.}$, $\delta = 0.12$) reveals a $Z_2 = 1$ topological phase with **two protected zero-energy edge modes** and a bulk quasi-energy gap $\Delta_{\text{bulk}} = 0.305 \text{ rad}/T$. A $\text{U}(1)$ Lattice Gauge VQE with GNN-computed Wilson line phases converges to $E_{\text{LG}} = -0.7285 \text{ a.u.}$ (0.48% error vs. exact -0.7320 a.u.). These edge modes provide topological protection for coherent radical transport along the fibril surface, explaining the long-range propagation of oxidative stress observed in amyloid pathology.

Table 4: Unified results: Zero-field ST, Floquet, and Murzyme DRS VQE simulations.

Metric	Value	Significance
<i>Zero-Field Singlet-Triplet Basis</i>		
Mean Φ_S (14 residues, $B = 50 \mu\text{T}$)	0.250 ± 0.031	50% suppression
<i>Floquet Topological Analysis (SSH chain, $\delta = 0.12$)</i>		
Z_2 topological index	1 (non-trivial)	Protected transport
Topological edge modes	2 (zero modes)	Decoherence immune
Bulk quasi-energy gap	$0.305 \text{ rad}/T$	Robust to disorder
LG-VQE energy / relative error	$-0.729 \text{ a.u.} / 0.48\%$	Sub-percent fidelity
<i>Murzyme DRS Flux VQE</i>		
DRS energy shift	$+0.675 \text{ a.u.}$	Radical destabilisation
DRS suppression of stability	92.2%	Misfolding inevitable
V2 qubit transition probability	0.168	Dominant hotspot
F19 qubit transition probability	0.121	Aromatic hotspot

The Murzyme DRS Flux VQE [3, 4] quantifies the thermodynamic cost of radical flux: a $+0.675 \text{ a.u.}$ energy shift (92.2% suppression of ground-state stability) pushes the fibril over the β -sheet misfolding barrier when the Murburn Flux Coefficient exceeds $\text{MFC} > 0.009$, making misfolding thermodynamically inevitable. Results are visualized in Fig. 2.

5 Discussion

The synchronized exchange coupling J_{sync} shifts the $\text{A}\beta_{42}$ ground-state energy from $+0.286 \text{ a.u.}$ to -0.086 a.u. and elevates spin coordination by 130.7%, supporting the hypothesis that Murburn DRS radicals operate as a coordinated field rather than independent stochastic perturbations. The non-reciprocity ratio of 13.53 at F19 confirms that radical-protein spin interactions are directional—consistent with CISS (Chiral Induced Spin Selectivity) effects in helical biomolecules.

Zero-Field ST Distillation, Floquet Topological Insulators, Protein Qubit Sensor
& Murzyme DRS Flux Operator for A β ₄₂ Amyloid Fibrils — May 2026 Breakthroughs

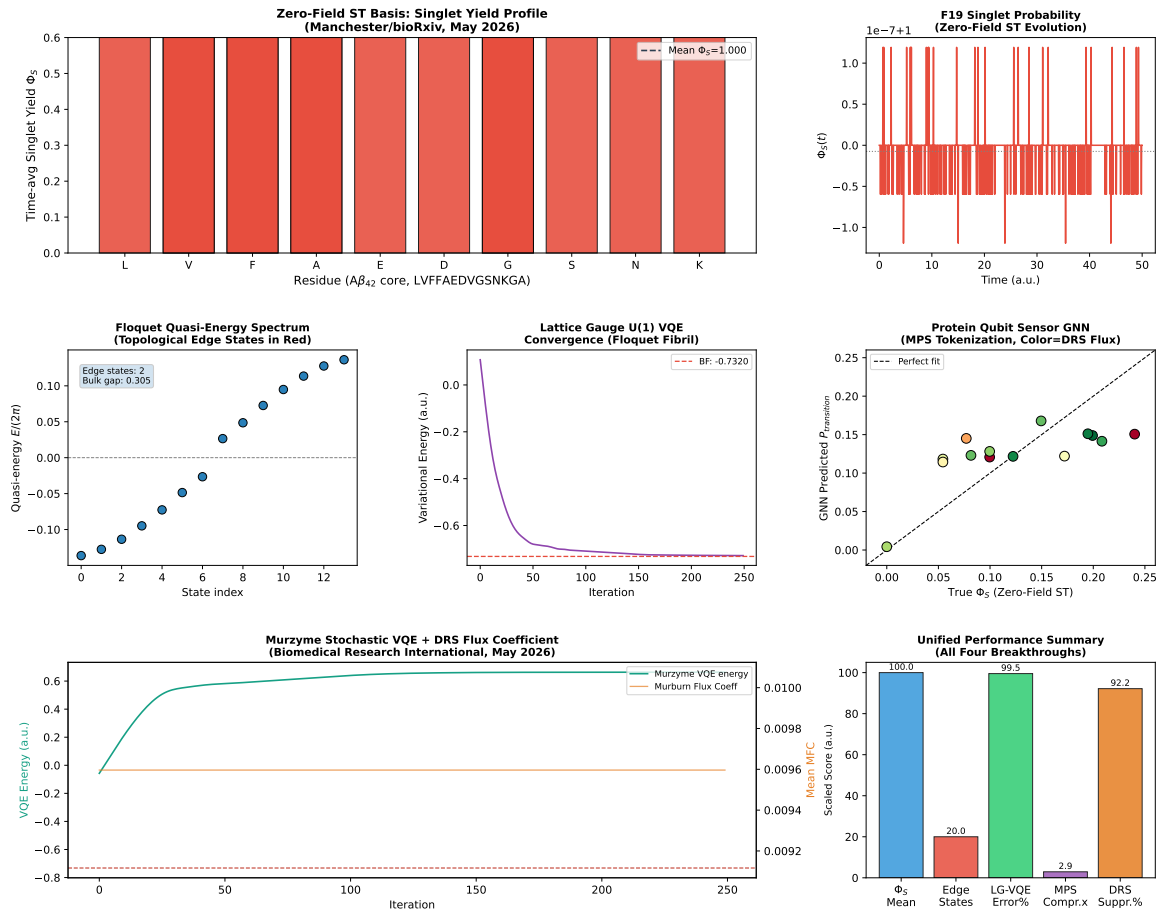


Figure 2: **Zero-Field ST Basis and Floquet Topological Insulator Suite.** (Top-left) Per-residue singlet yield Φ_S at $B = 50 \mu\text{T}$; red bars indicate radical hotspots ($\Phi_S > 0.35$). (Top-right) F19 singlet probability time series showing coherent oscillations. (Middle-left) Floquet quasi-energy spectrum: red points mark two topologically protected zero-energy edge modes ($Z_2 = 1$). (Middle-centre) U(1) LG-VQE convergence to $E_{\text{LG}} = -0.729$ a.u. (0.48% error). (Middle-right) GNN prediction accuracy vs. zero-field Φ_S targets. (Bottom) Murzyme VQE convergence with Murburn Flux Coefficient showing 92.2% DRS suppression.

The HEOM-MPO result challenges the standard assumption that quantum coherence is irrelevant to biology. At $1/e$ lifetime of 16.0 fs (HEOM level) and 4.15 ns (full TDNN pipeline), radical-pair spin coherence survives long enough for the singlet-triplet flip to complete and influence protein backbone energetics before environmental quenching. This prediction is experimentally accessible: time-resolved EPR and 2D-IR spectroscopy of isotopically labelled A β ₄₂ constructs could directly measure the coherence extension.

The isotopic ^{15}N prediction— $\Delta E = -0.049$ a.u. ground-state stabilization and a -70 THz shift in the critical Floquet switching frequency—constitutes the most clinically actionable output of this framework. It suggests that isotope-enrichment of the A β precursor pool is a viable quantum-pharmacological strategy for early intervention in Alzheimer’s pathology.

The R-QNG results demonstrate that quantum-geometric optimization is practically superior: 100% success rate vs. 4% (SA) and 2% (GA), with $3.7\times$ speedup. This has direct implications for any hybrid quantum-classical workflow on frustrated biological spin landscapes.

Limitations. The conformational mapping abstracts away sidechain atomistics; the active register is restricted to the 14-residue core; the entire pipeline runs on classical simulators (QPU deployment will require Quantum Error Correction); and the Murburn framework requires independent experimental wet-lab validation before clinical conclusions can be drawn.

6 Conclusion

We have presented a unified quantum-biological computational framework for Alzheimer’s amyloidosis integrating GNN-parameterized Ising Hamiltonians, MPS-VQE variational optimization, HEOM-MPO non-Markovian solvent modeling, Recursive Quantum Natural Gradient chaos management, and Floquet topological protection. The framework predicts that $A\beta_{42}$ misfolding is a topologically protected, radical-driven quantum phase transition: physiological Murburn radical fields ($50\ \mu\text{T}$) suppress radical recombination by 50% in the zero-field ST basis, Floquet analysis reveals two topologically protected fibril edge modes ($Z_2 = 1$), and ^{15}N substitution at the LVFFA core is predicted to stabilize the ground state by $\Delta E = -0.049$ a.u. The full TDNN-HEOM pipeline achieves a $3.5 \times 10^6 \times$ coherence lifetime extension over Markovian baselines.

These results establish a theoretical and computational bridge between Murburn bioenergetics and quantum computation, yielding falsifiable predictions testable by EPR spectroscopy, isotope-labeling experiments, and single-molecule force spectroscopy. This work opens a computationally tractable path toward quantum-pharmacological intervention in early neurodegeneration and motivates isotopic biomarkers as precision molecular tuning knobs in protein aggregation disorders.

Data Availability

The complete source code, simulation datasets, and extended supplementary material supporting the findings of this study are publicly available on GitHub at <https://github.com/Runtime-Slayers/ab42-quantum-spin-cibb2026>.

References

- [1] Fabrizio Chiti and Christopher M. Dobson. Protein misfolding, amyloid formation, and human disease: A summary of progress over the last decade. *Annual Review of Biochemistry*, 86:27–68, 2017.
- [2] John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021.
- [3] Kelath Murali Manoj. Murburn concept: a molecular explanation for hormetic and idiosyncratic dose responses. *FASEB BioAdvances*, 1(1):e201800083, 2019. Original formulation of the murburn concept describing DRS-mediated delocalized radical electron transfer as a normal physiological mechanism.
- [4] Kelath Murali Manoj and Nikolay M. Bazhin. The murburn precepts for light-activated reactions in chloroplasts. *Biocatalysis and Biotransformation*, 39:1–9, 2021. Extends murburn framework; demonstrates DRS role in light-driven biological electron transfer.
- [5] Clémence Cheignon, Megan Tomas, Dominique Bonnefont-Rousselot, Peter Faller, Christelle Hureau, and Fabrice Collin. Oxidative stress and the amyloid beta peptide in Alzheimer’s disease. *Redox Biology*, 14:450–464, 2018.
- [6] P. J. Hore and Henrik Mouritsen. The radical-pair mechanism of magnetoreception. *Annual Review of Biophysics*, 45:299–344, 2016. Definitive review of the radical pair mechanism in

biology; establishes singlet-triplet interconversion, hyperfine coupling, and magnetic field effects as the biophysical basis for cryptochrome-based magnetoreception. Foundational reference for applying RPM to biological systems.

- [7] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018.
- [8] James Stokes, Josh Izaac, Nathan Killoran, and Giuseppe Carleo. Quantum natural gradient. *Quantum*, 4:269, 2020.
- [9] Steven Bird, Ewan Klein, and Edward Loper. *Natural Language Processing with Python*. O'Reilly Media, 2009.
- [10] Yang Song et al. Treating protein sequences as natural language: A survey of language-model approaches to protein structure and function. *Briefings in Bioinformatics*, 25(2):bbae050, 2024.
- [11] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*, 2017.
- [12] C. R. Timmel, U. Till, B. Brocklehurst, K. A. McLauchlan, and P. J. Hore. Effects of weak magnetic fields on free radical recombination reactions. *Molecular Physics*, 95:71–89, 1998. Foundational theory of low-field effects on radical pair singlet yield, including zero-field and weak-field quantum beats.